Unsupervised Domain Adaptation for Cross-Scene Multispectral Point Cloud Classification

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***Abstract*— Remote sensing cross-scene classification has always been an important research field, especially in the field of 3-D classification, which is of great significance. Considering the diversity of collection conditions, seasons, and regional styles, deep learning networks well-trained on one source domain dataset tend to suffer from severe performance degradation when applied to other target domain datasets. To tackle the issue, in this article, we propose a new cross-scene classification method, which combines pre-alignment and Shannon entropy constraint to accomplish unsupervised domain adaptive classification (PS-UDA). On the one hand, the pre-alignment employs L_2-paradigm constraint and Laplace matrix to pre-align the features. With the L_2-paradigm constraint, the originally distant features of the source and target domain are constrained to the same sphere surface, and it is easier to make the distribution alignment on the sphere surface. Further, the Laplace matrix is used to map the source and target domain. In this way, similar features of the source and target domain are further aligned, and dissimilar features become discrete from each other. On the other hand, this article employs the Shannon entropy constraint to motivate the network to obtain more high-confidence target domain pseudo-labels. In addition, to fully utilize the unlabeled target domain information, the target domain features are augmented using the adjacency matrix. Experimental results of two cross-scene multispectral point cloud classifications demonstrate that the proposed PS-UDA can effectively mitigate the spectral shift issue in cross-scene multispectral point clouds, achieving state-of-the-art performance.**

***Index Terms*— Cross-scene classification, graph convolutional network (GCN), multispectral point cloud classification, pseudo-label, unsupervised domain adaptive (UDA).**

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NOMENCLATURE

Abbreviations

MS-LiDAR	Multispectral light detection and ranging.
SD	Source scene data.
TD	Target scene data.
DA	Domain adaptation.
GCN	Graph convolutional network.
UDA	Unsupervised domain adaptive.

Variables

$\mathbf{X}$	Normalized feature matrix.
$\mathbf{X}_s$ and $\mathbf{X}_t$	Source and target data.
$\mathbf{Z}_s$ and $\mathbf{Z}_t$	Features extracted by GCN.
$\mathbf{Y}$	Class labels.
N_s and N_t	Numbers of SD and TD points.
$\mathbf{A}$	Adjacency matrix.
$\mathbf{H}$	Shannon entropy.
C	Number of classes.
M	Feature dimension.
l	Number of feature channels.

I. INTRODUCTION

MS-LiDAR, as a new remote sensing technology, can collect 3-D point clouds with spectral information. The data collected by MS-LiDAR is called a multispectral point cloud. This novel technique extends the spectral imaging scope to stereoscopic spectral sensing. A finer scale 3-D classification of land cover can be achieved by only using a multispectral point cloud. Recent studies have also demonstrated the excellent performance of MS-LiDAR in real-scene applications [1], [2], [3], [4], [5].

There are currently three technical routes for multispectral point cloud classification: PointNet-based, CNN-based, Transformer-based, and GCN-based. The PointNet-based method is inefficient in handling large amounts of data due to its inherent limitation of computational complexity. The CNN-based method requires a large number of labeled samples to ensure training convergence [6]. Currently, due to hardware technology limitations, there is an insufficient number of training samples for multispectral point cloud classification to train such a large-scale model. Although powerful CNN models are available for pretraining. For multispectral point clouds, the strong coupling relationship between spectral

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TABLE I
PARAMETERS COMPARISON

Method	Parameters
Point Transformer V1	11.4M
Fast Point Transformer	37.9M
Point Transformer V2	12.8M
Point Transformer V3	46.2M
GCN	260K

and geometric information makes the voxelization or rasterization process lose both spectral and geometric information. The Transformer-based methods draw on the self-attention mechanism. Point Transformer V1 introduces a self-attentive layer for point clouds and this layer is naturally adapted to point cloud processing [7]. To improve its processing efficiency and model performance, scholars have conducted in-depth studies. Fast Point Transformer encodes continuous 3-D coordinates, and the voxel hashing-based architecture boosts computational efficiency [8]. Point Transformer V2 groups point clouds and introduces a finer attention mechanism [9]. Point Transformer V3 introduces grouped vector attention and partition-based pooling mechanisms to improve processing efficiency and model performance [10]. However, the structure of these methods is still complex compared to GCN, with more number of parameters, as shown in Table I. The GCN-based methods have relatively fewer parameters and allow the point cloud to be input directly without rasterization or voxelization. The first law of geography states that "Everything is related to everything else, but near things are more related to each other." For multispectral point cloud data, the spatial distribution of spectral features can reflect the land cover distribution information. GCN can represent the spectral feature spatial distribution by topological adjacency with its unique advantage. Therefore, scholars believe that graph theory can be used to model MS-LiDAR point clouds to adapt the unstructured and irregularly distributed features and have achieved outstanding achievements. Li et al. [1] proposed a GCN-based MS-LiDAR point cloud-building extraction framework, which only inputs the raw multispectral point cloud and directly outputs the point-by-point building extraction results. Zhao et al. [2] combined GCN and PointNet as well as proposed a feature inference module for feature augmentation. Wang et al. [3] proposed a multi-attribute smoothed GCN for multispectral point cloud classification. Li et al. [4] proposed attentive graph geometric moments convolution to efficiently extract spatial geometric structure features based on GCN. Wang et al. [5] introduced spectral graph aggregated information as residuals based on GCN, which effectively utilized the respective advantages of spatial and spectral features to improve the multispectral point cloud classification performance. The above methods require a large training dataset to achieve optimal performance. Whereas collecting and labeling large numbers of point clouds is often laborious and time-consuming. On the other hand, they use training and testing data sharing the same distribution, the cross-scene distribution shift makes these methods not work. Spectral shift often exists between MS-LiDAR point clouds collected in different scenes. This is because the spectral information of MS-LiDAR data will inevitably be affected by many factors such as seasonal and weather changes, resulting in differences in spectral reflectance between the same land cover in different scenes. Because the train data and test data used in the above methods are independently and identically distributed, spectral differences between different scenes tend to degrade the performance of the trained model when it is directly applied to other scenes. As a result, the current MS-LiDAR point classification methods are not directly transferable to other scenes and cannot be tested on unlabeled data collected in real-time. This has become a major constraint in MS-LiDAR data analysis and understanding.

Fortunately, the UDA technique can help mitigate this problem and has been successfully used in cross-scene classification. Early works based on UDA focused on aligning the edge distributions of the SD and TD data [11], [12]. Interdomain alignment by optimizing differences [13], [14] or using adversarial training in the input layer [15], [16], [17], [18], [19], intermediate feature layer [20], [21], [22] or output layer [23] to reduce distribution differences. However, traditional backbone networks (e.g., VGG [24], ResNet [25]) targeting image data are not applicable directly to point cloud data. Converting the 3-D point cloud to a 2-D feature map can result in a loss of information. GCN is suitable for point cloud data but is difficult to extract domain invariant features because of its shallow layers. Therefore, the alignment of point cloud feature distribution cannot be relied on by graph neural networks alone.

Recently, the proposition of pseudo-labels has instilled hope for achieving high-precision cross-scene multispectral point cloud classification. Networks are trained using the most reliable pseudo-labels [26], [27] extracted from the target domain data. Consequently, the accuracy of pseudo-labels assumes paramount importance in the quality of TD classifiers [29]. Li et al. [30] augmented pseudo-label generation by clustering targets based on geometric resemblance. However, for large-scale remote sensing point clouds, the domain shift induced by direct scene replacement exacerbates significantly, rendering reliance solely on geometric similarity insufficient. Additionally, Deng et al. [31] proposed a hierarchical contrast adaptive method to encourage classifiers to learn domain-invariant but differentiated features. However, the fixed confidence threshold limits pseudo-label extraction for difficult samples. Vu et al. [32] explored the potential of confidence by introducing entropy loss to directly penalize low-confidence predictions in the target domain, pioneering the application of entropy-based UDA training to semantic segmentation tasks. Nevertheless, they overlook the errors and noise inherent in pseudo-labels. Addressing this concern, Zhang et al. [33] resorted to online denoising prototypes to enhance performance. Li et al. [34] introduced a distribution alignment technique to synchronize the class distributions of clustering assignments with those of pseudo-labels. Wang and Zhang [35] proposed leveraging intraclass similarity within the TD to rectify inaccurately generated pseudo-labels. Yuan et al. [36] advocate for a progressive alignment strategy

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that minimizes reliance on pseudo-label supervision. However, pseudo-label generation followed by correction may not adequately meet real-time requirements. Mattolin et al. [37] suggested progressively filtering pseudo-labels based on confidence levels varying from lenient to stringent during training. In the case of too much variation between domains, the network may be overconfident leading to false predictions with high confidence. Biswas and Tešić [38] endeavored to investigate contrastive learning concerning both local and global image features with the aim of effecting feature alignment across task-specific layers. However, within the framework of contrastive learning, the selection of false negative examples can introduce disruptions to the learning process and detrimentally impact overall performance. In response to this challenge, they introduced a support-guided debiased contrastive learning approach for DA, aimed at appropriately labeling the unlabeled target dataset and mitigating bias toward target detection. Moreover, they introduced a novel support-guided progressive pseudo-labeling method for generating instance labels within the target domain [39].

Although all the aforementioned methods perform very well, there is an obvious problem. The neural networks designed by the above methods are usually deep networks. If the GCN is directly applied to the above network, it will cause performance degradation due to its shallow layers. There are currently graph theory-based point cloud cross-domain recognitions. For instance, Wang et al. [40] utilized DGCNN for cross-domain recognition of 3-D objects. They focus on the geometric structure of different classes of objects to assist the network in extracting domain-invariant features. However, there are many areas with similar geometrical structures but different classes in the remotely sensed scene. For example, the geometry of grass and bare ground is very similar. Park et al. [41] utilized graph convolution to learn the implicit shape information of the source domain. However, for remotely sensed scenes, it is challenging to use only shape information to assist cross-domain classification. As a result, it is difficult for GCN to obtain a large number of high-precision pseudo-labels for the target scene. In addition, in the absence of alignment, most of the predicted target domain classification labels obtained are wrong, and there are no truth labels available for the target domain. It is difficult to filter out those pseudo-labels that can be used for subsequent training. These problems make cross-scene multispectral point cloud classification more challenging. As a result, we can conclude that the pre-alignment step is very important. This could make up for the shortcomings of GCN.

For better pre-alignment, we propose to use the L_2-paradigm constraint and Laplace matrix to align multispectral point cloud feature distributions between different scenes. This alignment has two main advantages. On the one hand, the scale or magnitude of different scene data can vary greatly. The L_2-paradigm constraint allows different scene data to be mapped in the same spherical space. This eliminates data discrepancies and makes it uniform in scale. On the other hand, since the L_2-paradigm constraint only aligns features globally, we propose to use the Laplace matrix to align similar features in spherical space. For illustrative purposes, Figs. 1–3 provide the original feature distribution map, the feature distribution map after the L_2-paradigm constraint, and the feature distribution map after the Laplace matrix alignment.

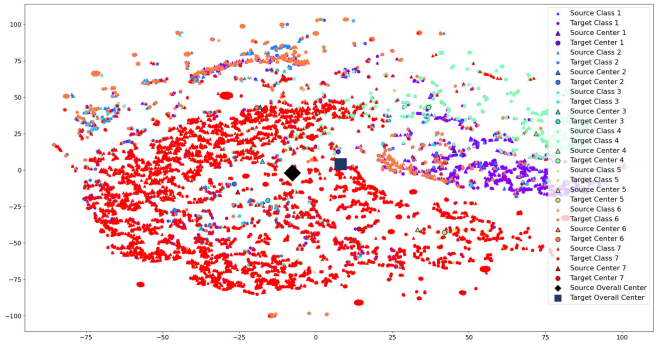


Fig. 1. Original feature distribution map.

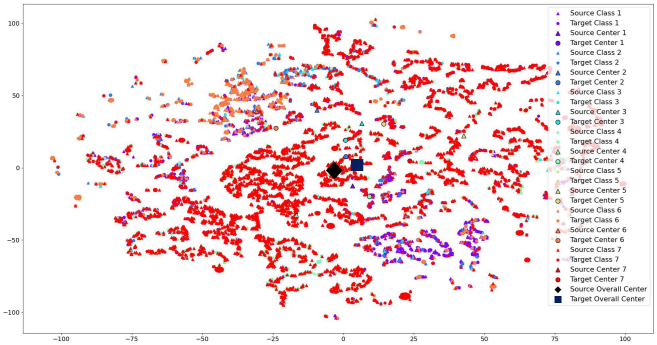


Fig. 2. Feature distribution map after the L_2-paradigm constraint.

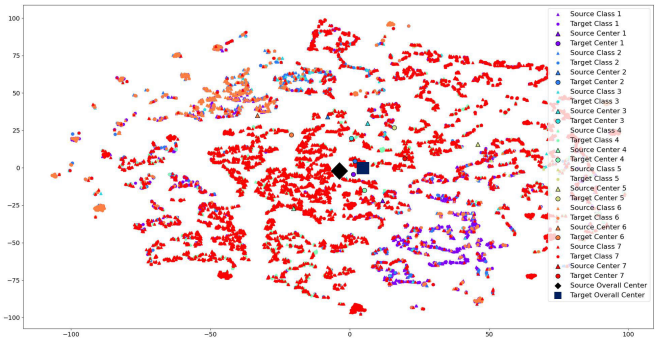


Fig. 3. Feature distribution map after the Laplace matrix alignment.

The feature distributions of different scenes are aligned on the whole and the distances between different classes are reduced. This relieves the pressure on aligning features by GCN. However, since the features of the same class are not fully aligned, the target domain does not provide any labels during training. This poses a challenge to predict a large number of high-confidence pseudo-labels. Therefore, it is important to further constrain pseudo-labeling. The currently popular pseudo-label screening strategies usually screen pseudo-labels based on confidence ranking. For this reason, we propose to constrain the network according to the confidence level. Specifically, we propose to use the Shannon entropy [42] to measure the determination of the network

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for prediction. Using this measure as a regularization term in the loss function promotes high-confidence pseudo-label predictions. In this way, the number of pseudo-labels that can be screened out will increase. The filtered pseudo-labels are used to train the target domain training and finalize the cross-scene classification.

In this article, we propose to pre-align the source and target scene MS-LiDAR point clouds to alleviate the lack of GCN alignment feature capability and constrain the network to improve the quality of the pseudo-labels. To the best of our knowledge, we are the first to employ the UDA technique to mitigate the effects of spectral shifts in the MS-LiDAR point clouds. To avoid the situation where no pseudo-labels are screened for some of the difficult classes when fixed confidence thresholds are screened for pseudo-labels. We introduce a straightforward and innovative approach for screening pseudo-labels, assessing them based on their percentile rankings. In contrast to existing graph-based multispectral point cloud processing models, our proposed PS-UDA can effectively mitigate the performance degradation that occurs when there are distributional differences between the Test Set and the Train Set. The main contributions can be summarized as follows:

1) We introduce the L_2-paradigm and Laplace matrix to pre-align the features of the source and the target MS-LiDAR point cloud. In this way, the same land cover spectral features become more similar. The number of available pseudo-labels of the target MS-LiDAR point cloud can be significantly improved by the pre-alignment, which is helpful for subsequent training.
2) We introduce the Shannon Entropy constraint to penalize the network for low-confidence predictions in TD. So, more high-confidence pseudo-labels are obtained for target network training to classify TD.

At this moment, it is important to reiterate that the key innovation of the proposed method in this article lies in the fact that the whole process does not require any real labels of the target domain, and the unfamiliar scene (without real labels) can be predicted based on the known scene directly, which mitigates the effect of spectral shift. The remainder of this article is organized as follows. Section II describes the foundation part of the article. Section III describes the proposed PS-UDA in detail. Section IV evaluates the proposed method via experiments with two real multispectral point cloud datasets. Section V concludes this article with some remarks and hints at plausible future research lines.

II. Preliminaries

Some techniques used in the proposed method are described in this section. Define that $P_s = \{p_s^{(i)}\}_{i=1}^{N_s}$ and $P_t = \{p_t^{(i)}\}_{i=1}^{N_t}$ are the MS-LiDAR points of SD and TD, respectively. $\mathbf{Y_s} = \{y_s^{(i)}\}_{i=1}^{N_s}$ are the class labels $\{1, 2, \ldots, C\}$ from SD, where C is the number of classes. Note that, there are no labels of TD in the unsupervised case. Here, N_s and N_t denote the number of source points and the number of target points, respectively. Some important abbreviations and variables and their descriptions are given in Nomenclature.

A. Graph Construction

Because of the large size of the MS-LiDAR point cloud, the point cloud is segmented into superpoints using the method proposed in our previous work [43]. In this article, uniform superpoints will be used as input data for the comparison methods. Please note that each dimension of the superpoint's feature vector represents the mean of the corresponding dimension's features across all points within the superpoint. In addition, using the K-Nearest Neighbors (KNN) algorithm, each superpoint is constructed as a node in a graph based on the spatial and spectral characteristics.

B. Maximum Mean Discrepancy (MMD)

Among the existing methods, MMD is an effective non-parametric distance metric. It calculates the mean discrepancy between the SD and TD in the reproducing kernel Hilbert space. The MMD distance quantifies the dissimilarity between two distinct domains. Furthermore, the MMD distance of the SD and TD is described as

$$\mathrm{MMD}(\mathbf{X_s}, \mathbf{X}_t) = \left\| \frac{1}{N_s} \sum_{i=1}^{N_s} \phi((\mathbf{x}_s)_i) - \frac{1}{N_t} \sum_{j=1}^{N_t} \phi((\mathbf{x}_t)_i) \right\|_H^2 \quad (1)$$

where ϕ is the mapping function to map the original variables into the higher dimensional space.

Based on the samples of two distributions, the mean value of the samples of different distributions on the mapping function ϕ is determined by finding the mapping function ϕ on the sample space. The MMD is obtained by finding the ϕ such that mean discrepancy has a maximum value, and finally, the MMD is taken as the test statistic to determine whether the two distributions are the same. If this value is small enough, the two distributions are considered the same, otherwise they are considered different. A simpler way to understand this is to find the distance between the means of two stacks of data in a high-dimensional space.

III. Methodology

The flowchart of the proposed PS-UDA is shown in Fig. 4, which consists of two stages. The first stage is to generate the crude pseudo-label. The second stage is TD multispectral point cloud classification. Specifically, the features of the two domains are pre-aligned at the beginning. To extract features with rich semantic information, the two-domain feature matrix and adjacency matrix are fed into a GCN. The GCN processes the feature matrix and adjacency matrix to generate feature representations for both SD and TD, denoted as $\mathbf{Z}_s$ and $\mathbf{Z}_t$ respectively. After the first stage, the pseudo-labels of the TD are obtained. Then, the high-quality pseudo-labels are screened according to the confidence level and used for the TD-GCN training. Before training the TD-GCN classifier, the feature matrix and adjacency matrix of the TD are spliced as the input features of the TD-GCN classifier. Finally, the TD-GCN is used to classify the TD multispectral point cloud. It is worth noting that our proposed method is an end-to-end network, and the TD is unlabeled.

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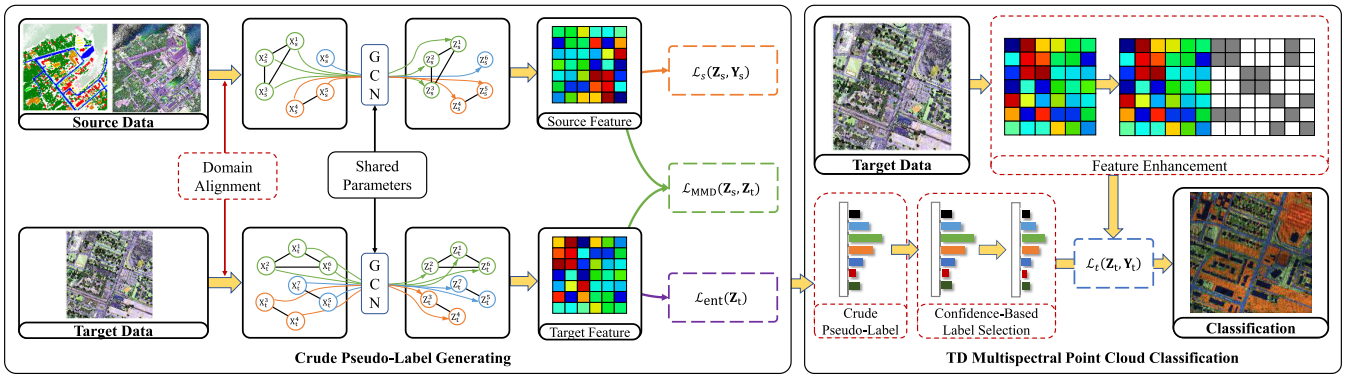


Fig. 4. Flowchart of the proposed PS-UDA, including crude pseudo-label generating and TD multispectral point cloud classification.

A. Crude Pseudo-Label Generating

The main task in this stage is to obtain high-quality TD pseudo-labels to ensure the effectiveness of the next stage of training. This stage consists of three components: 1) Domain Alignment; 2) Minimum Entropy Constraint; and 3) Classifier of GCN.

1) Domain Alignment: When the difference between the SD and TD is relatively significant, the feature distributions of the two domains may be very different. This makes it difficult for the network to obtain TD high-quality pseudo-labels, and it is not possible to make feature alignment directly. Therefore, each feature of the feature matrix $\mathbf{X}$ can be constrained by the L_2 paradigm constraint: $\widetilde{x} \leftarrow (\widetilde{x}/\|\widetilde{x}\|_2)$. The originally distant features of SD and TD are constrained to the same sphere surface, and it is easier to make the distribution alignment of SD and TD on the sphere surface. With the L_2 paradigm constraint, the distributions between SD and TD have been brought closer compared to the original. Further, inspired by spectral clustering, the Laplace matrix $\mathbf{L}$ is used to map the SD and TD. In this way, the feature distributions of SD and TD are further aligned. Specifically, the $\mathbf{X}_s \in \mathbb{R}^{N_s \times M}$ and $\mathbf{X}_t \in \mathbb{R}^{N_t \times M}$ are concatenated to obtain the overall feature matrix $\mathbf{X} \in \mathbb{R}^{(N_s+N_t)\times M}$. The similarity matrix $\mathbf{W}$ is calculated from the overall feature matrix $\mathbf{X} \in \mathbb{R}^{(N_s+N_t)\times M}$. The diagonal matrix $\mathbf{D}$ is further calculated according to the following:

$$\mathbf{D}_{ii} = \sum_j \mathbf{W}_{ij}. \tag{2}$$

Therefore, the Laplace matrix $\mathbf{L}$ is obtained according to $\mathbf{L} = \mathbf{D} - \mathbf{W}$. Further, the feature matrix $\mathbf{X}$ is updated to $\widetilde{\mathbf{X}} = \mathbf{L}^{\mathbf{T}}\mathbf{X}$.

2) Minimum Entropy Constraint: After performing the initial feature alignment, there is still uncertainty about the classification of the target domain. Vu et al. [32] showed the advantage of reducing the entropy of the target domain. Inspired by [32], we hope to introduce entropy to constrain the uncertain classification of the network to the target domain and increase the number of high-confidence pseudo-labels. The Shannon entropy [42] of each point of TD is calculated according to (3). Given the points P_t of TD, the entropy map $\mathbf{H}_{\boldsymbol{x}_t} \in [0, 1]^{N_t}$ is composed of the independent point-wise entropies normalized to [0, 1] range

$$\mathbf{H}_{\boldsymbol{x}_t}^{N_t} = \frac{-1}{\log(l)} \sum_{l=1}^{l} \mathbf{P}_{\mathbf{x}_t}^{N_t} \log \mathbf{P}_{\boldsymbol{x}_t}^{N_t} \tag{3}$$

where $\mathbf{P} \in \mathbb{R}^C$ is the prediction vector and l is the number of feature channels of the multispectral point cloud. The entropy loss $\mathcal{L}_{\mathrm{ent}}$ is defined as the sum of all point-wise normalized entropies

$$\mathcal{L}_{\mathrm{ent}}(P_t) = \sum \mathrm{H}_{\boldsymbol{x}_t}^{N_t}. \tag{4}$$

The following is an example of a binary classification problem: assuming that there are two classes of points in the TD, and after the GCN, a two-channel semantic segmentation graph is finally obtained, which can be represented by the one-hot vector ($\mathbf{x}$, $\mathbf{y}$). When $\mathbf{x}$ or $\mathbf{y}$ tends to 1, it indicates that the network can determine the class of the point, and the $\mathbf{H}$ is small. Conversely, when the difference between $\mathbf{x}$ and $\mathbf{y}$ is so small the network is hard to determine the class of the point, and the $\mathbf{H}$ is high. Therefore, to augment the confidence of the pseudo-labels in TD, the entropy loss $\mathcal{L}_{\mathrm{ent}}$ is used to directly penalize the low confidence prediction of TD to increase the number of available pseudo-labels of TD.

3) Classifier of GCN: The cross-entropy loss is commonly used for classification problems. It is the key metric used to optimize model parameters, measuring the difference between the actual results and the predicted results of the model. The cross-entropy loss of the labeled SD can be defined as follows:

$$\mathcal{L}_s(\mathbf{Z}_s, \mathbf{Y}_s) = -\frac{1}{N_s} \sum_{i=1}^{N_s} y_s^i \log\left(\hat{y}_s^i\right) \tag{5}$$

where y_s^i is the truth label of the ith source scene node, and $\hat{y}_s^i$ is the predictive label of the ith source scene node.

Taking into account the aforementioned loss functions, the total loss can be expressed as follows:

$$\mathcal{L}_{\mathrm{total}}(\mathbf{Z}_s, \mathbf{Z}_t, \mathbf{Y}_s) = \mathcal{L}_s(\mathbf{Z}_s, \mathbf{Y}_s) + \lambda_1 \mathcal{L}_{\mathrm{MMD}}(\mathbf{Z}_s, \mathbf{Z}_t) + \frac{\lambda_2}{N_t} \mathcal{L}_{\mathrm{ent}}(\mathbf{Z}_t) \tag{6}$$

where λ_1 and λ_2 are the hyperparameter which controls the contribution of $\mathcal{L}_{\mathrm{MMD}}$(refer to (1) for details) and $\mathcal{L}_{\mathrm{ent}}$.

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B. TD Multispectral Point Cloud Classification

In Section III-A, the network cursorily classifies the multispectral point cloud of TD. Obviously, it is impractical to use all the pseudo-labels for supervising the second-stage network training. Therefore, screening pseudo-labels for supervised second-stage network training and sufficiently utilizing the TD information is the core task of this stage. This stage consists of three components: 1) Confidence-Based Label Selection; 2) TD Feature Augmentation; and 3) Classifier of TD-GCN.

1) Confidence-Based Label Selection: To screen the pseudo-labels involved in training, an intuitive method is to set a high and fixed threshold τ to select those pseudo-labels with high confidence to compute the loss. While this strategy can make sure that only high-quality pseudo-labels contribute to the network training, when the difference between the SD and TD is relatively large, only a few pseudo-labels are selected. As a result, some of the accurate pseudo-labels will not be used for subsequent training. For classification tasks, the learning difficulty varies from different class. The easy classes, naturally, contain higher confidence levels and are more susceptible to being selected τ. While the difficult classes, with low confidence, are less susceptible to being selected. Therefore, for each class in the pseudo-label set $\mathbf{Y}_t = \{\hat{y}_t^{(i)}\}_{i=1}^{N_t}$, the top $\alpha\%$ pseudo-labels are selected manually based on their confidence level to supervise the second stage of network training. Therefore, the subsequent label set is updated to

$$\mathbf{Y}_t = \left\{\hat{y}_t^{(i)}\right\}_{i=1}^{N_t} \rightarrow \left\{\hat{y}_t^{(i)}\right\}_{i=1}^{N_t \times \alpha\%}. \tag{7}$$

2) TD Feature Augmentation: In graph theory, the adjacency matrix is a data representation used to represent the connection relationships between nodes in a graph. Therefore, to make full use of the information of TD, the adjacency matrix $\mathbf{A}_t \in \mathbb{R}^{N_t \times N_t}$ can be considered as a location feature. Further, the original feature matrix $\mathbf{X}_t \in \mathbb{R}^{N_t \times M}$ is concatenated with the adjacency matrix to obtain the augmented feature matrix $\mathbf{X}_t \in \mathbb{R}^{N_t \times (M+N_t)}$.

3) Classifier of TD-GCN: After obtaining the pseudo-labels and the feature matrix for training the second-stage network, a new GCN is trained on TD is notated as TD-GCN. The cross-entropy loss is used in TD-GCN when predicting the probability of TD and using the pseudo-labels to calculate the error. The cross-entropy loss of the TD is defined as

$$\mathcal{L}_t(\mathbf{Z}_t, \mathbf{Y}_t) = -\frac{1}{N_t \times \alpha\%} \sum_{i=1}^{N_t \times \alpha\%} \hat{y}_t^i \log\left(\widehat{\hat{y}}_t^i\right) \tag{8}$$

where $\hat{y}_t^i$ is the pseudo-label of the ith target scene node, and $\widehat{\hat{y}}_t^i$ is the predictive label of the ith target scene node.

Algorithm 1 shows the training process of our proposed PS-UDA. It is worth noting that we directly take the whole scene as input and do not need to split the scene into multiple sub-scenes as batches.

Algorithm 1 PS-UDA Training

Input:

Source LiDAR points $\mathbf{X}_s = \{x_s^1, \ldots, x_s^i\}$; Source LiDAR point labels $\mathbf{Y}_s = \{y_s^1, \ldots y_s^i\}$; Target LiDAR points $\mathbf{X}_t = \{x_t^1, \ldots, x_t^i\}$; Numbers of SD and TD points N_s, N_t; GCN_s.params initialization; GCN_t.params initialization

Output:

The prediction labels of target domain multispectral point cloud.

$\mathbf{X} = \mathbf{concat}(\mathbf{X}_s, \mathbf{X}_t)$;
$\mathbf{X} \leftarrow \mathbf{X} / \|\mathbf{X}\|_2$
Calculate similarity matrix $\mathbf{W}_{ij} = \begin{cases} 0 \text{ if } \exp(-\|x_i - x_j\|^2) < 0.7 \\ 1 \text{ other} \end{cases}$;
Calculate diagonal matrix $\mathbf{D}_{ii} = \sum_j \mathbf{W}_{ij}$;
Calculate laplace matrix $\mathbf{L} = \mathbf{D} - \mathbf{W}$;
$\mathbf{X} \leftarrow \mathbf{L}^{\mathbf{T}}\mathbf{X}$
$\mathbf{X}_s = \{x_1, \ldots x_{N_s}\}$, $\mathbf{X}_t = \{x_{N_s+1}, \ldots x_{N_t}\}$;
da_model.params = GCN_s.params;
for i in epoches_1 **do**
  Forward $\mathbf{X}_s$ and $\mathbf{X}_t$ to obtain predictive labels $\hat{y}_s$, $\hat{y}_t$;
  Obtain features extracted by GCN $\mathbf{Z}_s$, $\mathbf{Z}_t$;
  Obtain confidence $\mathbf{H}$ based on pseudo-label $\hat{y}_t$;
  $\mathcal{L}_s \leftarrow y_s, \hat{y}_s$
  $\mathcal{L}_{\text{MMD}} \leftarrow \mathbf{Z}_s, \mathbf{Z}_t$
  $\mathcal{L}_{\text{ent}} \leftarrow \mathbf{H}$
  Calculate loss as Eq.(6);
  loss.backward();
  da_model.update();
end for
for all $\hat{y}_t$ **do**
  for j in Number of classes C **do**
    if $\hat{y}_t$ is class j:
      $\hat{y}_t \leftarrow \hat{y}_{t_j}$
    end if
    Descending order $\{\hat{y}_{t_j}^{(1)}, \ldots \hat{y}_{t_j}^{(n)}\}$
    $\mathbf{Y}_{t_j} = \left\{\hat{y}_{t_i}^{(i)}\right\}_{i=1}^{n \times \alpha\%}$
  end for
  $\mathbf{Y}_t \leftarrow \mathbf{concat}(\mathbf{Y}_{t_1}, \ldots \mathbf{Y}_{t_j})$
end for
target_model.params = GCN_t.params
for i in epoches_2 **do**
  Forward $\mathbf{X}_t$ to obtain predictive labels $\widehat{\hat{y}}_t$;
  $\mathcal{L}_t \leftarrow \hat{y}_t, \widehat{\hat{y}}_t$
  loss.backward();
  target_model.update();
end for

IV. Experiments Results and Discussion

A. Dataset Description

The original point cloud data is collected by Titan and published by Optech. The Titan MS-LiDAR system is the first operational multispectral laser scanner with three independent wavelength active beams, i.e., 532, 1064, and 1550 nm. More detailed information about Titan MS-LiDAR can be found

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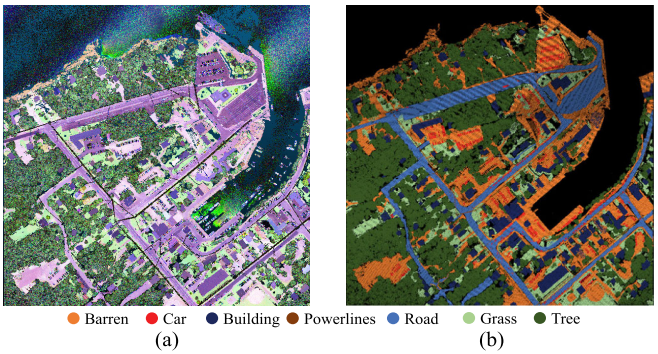


Fig. 5. HT dataset. (a) Multispectral point cloud. (b) Ground Truth.

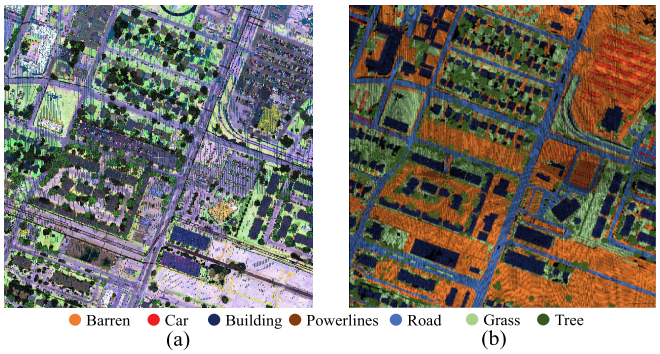


Fig. 6. UH dataset. (a) Multispectral point cloud. (b) Ground Truth.

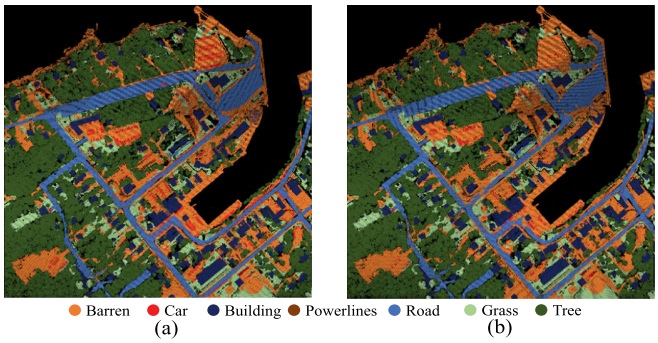


Fig. 7. Comparison of the original HT MS-LiDAR point cloud with the superpoints. (a) Ground Truth. (b) Superpoints.

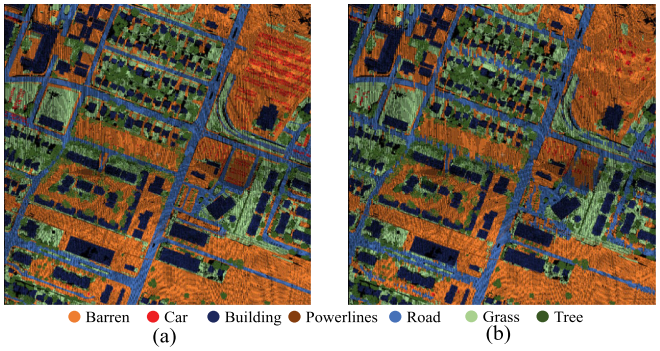


Fig. 8. Comparison of the original UH MS-LiDAR point cloud with the superpoints. (a) Ground Truth. (b) Superpoints.

in [44]. The data consists of three separate point clouds, each corresponding to a laser wavelength. The 1550 nm channel is used as the reference, which is matched with the nearest points in the other two channels by spatial nearest neighbor search. Then, assign the intensity values of the points to the nearest 1550 nm channel points. By the above steps, we obtained MS-LiDAR point cloud datasets with three wavelength intensity values: $[x, y, z, LRI_{\lambda 1550}, LRI_{\lambda 1064}, LRI_{\lambda 532}]$, which are obtained by processing the experimental data collected from the two study areas separately. The two datasets are described in detail as follows.

1) The first dataset is the Harbor of Tobermory (HT), which covers a port located in Tobermory with dimensions of 600 × 600 m. These data are collected via a vehicle at an average flight altitude of 500 m. This area contained 7 181 982 points. We have classified them into seven classes. It can be seen that water is originally available in the dataset, but this class is not available in the other dataset, so we do not use water as an input. Fig. 5 shows the multispectral point cloud and ground truth.
2) The second dataset is the University of Houston (UH), which is publicly available at the 2018 IEEE GRSS Data Fusion Contest. These data are also collected via vehicle at an average flight altitude of 500 m. A 595 × 600 m area of the UH Campus is used for the experiments in this article, and this area comprised 4 436 470 points. We have classified them into seven classes. Fig. 6 shows the multispectral point cloud and ground truth.

B. Implementation Details

All the networks in this article are implemented on the Pytorch platform. The equipment used in the following experiments consists of an NVIDIA Tesla V100 GPU with 16 GB of memory. We chose GCN as the backbone, its number of layers is set to 2, the number of hidden cells is set to 16, and the dropout is set to 0.2. The adaptive moment estimation (Adam) is chosen as the optimizer, whose initial learning rate is set with 1e − 3, and the training iteration is set with 5000. In the experiment, the point cloud data of SD and TD are respectively segmented into 8000 superpoints using the method proposed in our previous work [43]. Because each superpoint contains several points, the superpoints are relabeled by maximum voting so that they participate in the training as true labels. The predicted labels for superpoints will be mapped to all points inside the superpoint. Assuming correct prediction for all superpoints, the final semantic segmentation maps are shown in Figs. 7 and 8. It is evident that some of the points will be assigned wrong labels due to superpoint segmentation, which is a technical bottleneck in directly processing large-scale scenes based on superpoints, and negatively affects both the comparison participating methods and the proposed method, which will be further analyzed in the analysis of experimental results.

C. Evaluation Metrics

To better evaluate the participating comparison methods and the proposed PS-UDA, the intersection over union (IoU), the mean intersection over union (MIoU), the overall

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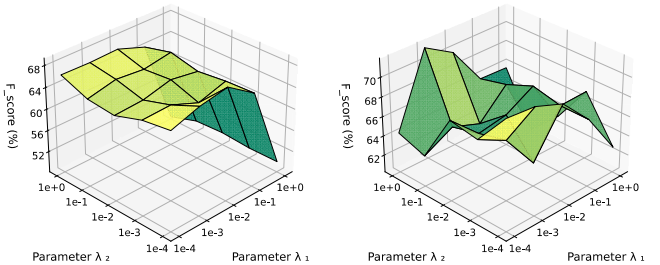


Fig. 9. Parameter tuning of λ_1 and λ_2 for the proposed PS-UDA using two datasets. (a) HT to UH. (b) UH to HT.

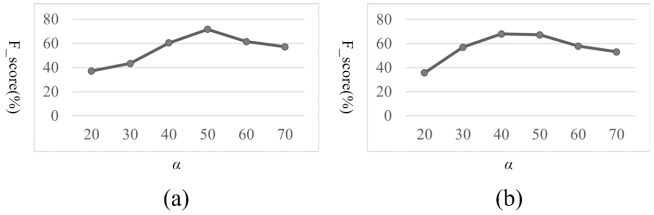


Fig. 10. Parameter tuning of α for the proposed PS-UDA using two datasets. (a) UH to HT. (b) HT to UH.

accuracy (OA), and the $F1$_score are employed to evaluate the classification performance. Since superpoints are used as input data, we validated the results by the superpoint-based within the superpoints.

D. Parameter Sensitivity Analysis

1) Parameter Tuning: In the proposed PS-UDA, the adjustable parameters are the regularization parameter λ_1, λ_2 and the reference ratio α for pseudo-label selection. The value range of λ_1, λ_2 are {1e − 4, 1e − 3, 1e − 2, 1e − 1, 1e + 0}. λ_1 and λ_2 are the important hyperparameters of PS-UDA, which relate to the quality of the pseudo-labels obtained in the first stage of the network. Fig. 9 shows the trend of the $F1$_score. In detail, the superpoints are labeled by the maximum voting method based on the class labels of the score of PS-UDA on the different SD datasets with different combinations of λ_1 and λ_2. To alleviate the pressure of parameter optimization, we select suboptimal parameters for analysis. It can be seen from Fig. 9 that parameter λ_1 in the range 1e − 3 to 1e − 4 achieves suboptimal performance.

This indicates that PS-UDA is less sensitive to the parameter λ_1 and λ_2 has its own suboptimal range in the two datasets. The above analysis confirms that the proposed Shannon entropy can play an important role in predicting the pseudo-labeling process. Therefore, HT as SD corresponds to suboptimal parameters λ_1 and λ_2 of 1e − 4 and 1e − 2; UH as SD corresponds to suboptimal parameters λ_1 and λ_2 of 1e − 4 and 1e − 3.

Since α is the threshold for subsequent pseudo-label screening, the value of α can directly affect the classification performance in the second stage. Fig. 10 shows the classification results with different α values on two cross-scene classification tasks in which F_score is used as the evaluation metric. From Fig. 10, it is evident that PS-UDA exhibits different performance for different thresholds of pseudo-labels. This is due to the fact that some of the difficult classes have fewer sample points, and once the threshold becomes high fewer supervised labels are achieved, while a threshold that becomes low introduces many erroneous pseudo-labels to participate in the subsequent training. Therefore, the hyperparameter α is set to 50, when the TD is HT; when the TD is UH, α is set to 40.

2) Ablation Study: The contributions of the components designed in PS-UDA are evaluated. Reviewing the entire network, PS-UDA implements cross-scene MS-LiDAR point cloud classification through information extraction at the pseudo-label level and feature level. Shannon entropy is used as a constraint term of loss to augment the quality of pseudo-labels. At the feature level, the L_2 paradigm constraint, and Laplace matrix $\mathbf{L}$ are used for feature pre-alignment and then the adjacency matrix is used for feature augmentation of TD features. Ablation analysis is performed to demonstrate the effectiveness of each module. Ablation experiments are first conducted at the pseudo-label level to verify the effectiveness of Shannon entropy. As shown in Table II, the introduction of Shannon entropy improves the performance of the network by a modest amount. This reflects the fact that Shannon entropy can augment the quality of pseudo-label and help with subsequent training.

TABLE II
ABLATION COMPARISON OF SHANNON ENTROPY

Dataset	PS-UDA (no Shannon entropy)	PS-UDA
HT → UH	F1_score :66.59%	F1_score :**68.01%**
UH → HT	F1_score :69.33%	F1_score :**71.66%**

Based on the retention of pseudo-labels, the validity of each module at the feature level is verified. As shown in Table III, the performance of the network without any module is significantly lower than that of PS-UDA. This is due to the limited ability of GCN to align SD and TD features. It can be seen that the introduction of the Laplace matrix $\mathbf{L}$ leads to a performance improvement of close to 8%. Although the addition of the L_2 paradigm constraint does not provide a significant advantage, the joint introduction of the L_2 paradigm constraint and Laplace matrix $\mathbf{L}$ can provide a significant advantage to the network. This reflects that pre-alignment can help the network classify the TD point cloud. Further, the performance of the network performing only feature pre-alignment is significantly lower than that of PS-UDA, which demonstrates that the adjacency matrix can be used as a location feature to augment TD features.

E. Performance on Cross-Scene MS-LiDAR Classification

Since there is still no cross-scene land cover classification method for multispectral LiDAR data, to better evaluate the performance of the PS-UDA method, we chose several representative methods. The methods involved in the comparison include a concentration of classical and advanced UDA algorithms, including k-means clustering algorithm, support vector machine (SVM) [45], GCN [46], which is the backbone used in this article, DSGCN-ASR [5], DANN [47], ProDA [33],

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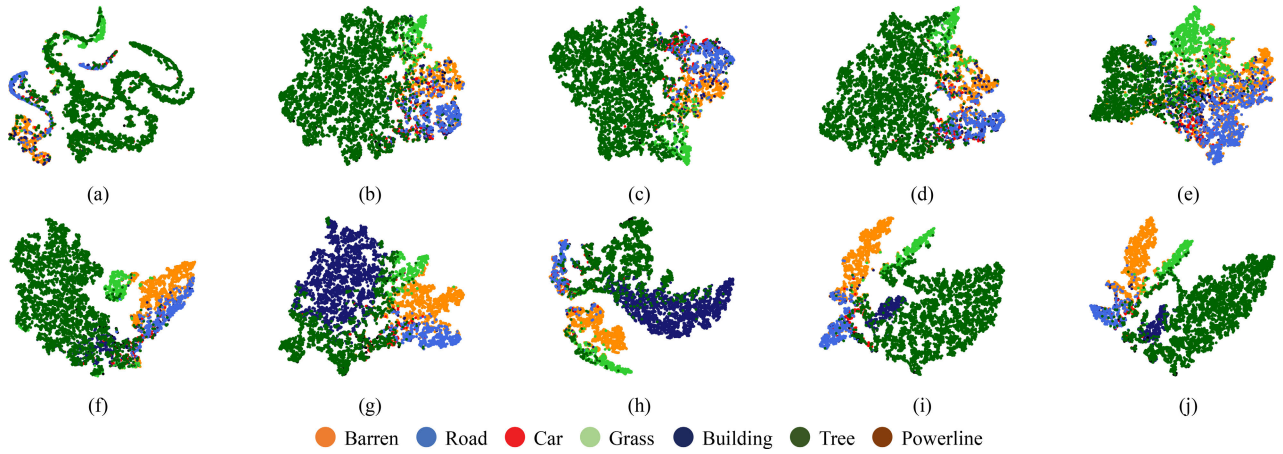


Fig. 11. Visualization of the unsupervised DA using t-SNE (The source domain is UH, and the target domain is HT). (a) SVM. (b) GCN. (c) DSGCN-ASR. (d) DANN. (e) SqueezeSeg V2. (f) ProDA. (g) HCA. (h) ConfMix. (i) TSRP. (j) PS-UDA (ours).

TABLE III
ABLATION COMPARISON OF MODULES AT FEATURE LEVEL

Dataset	L_2 paradigm constraint	Laplace	Feature augmentation	F1_score(%)
HT ↓ UH				51.38
	✓			52.29
	✓	✓		61.63
	✓		✓	63.11
		✓		59.86
		✓	✓	66.32
			✓	62.35
	✓	✓	✓	**68.01**
UH ↓ HT				54.40
	✓			55.00
	✓	✓		65.39
	✓		✓	67.70
		✓		61.78
		✓	✓	68.14
			✓	66.64
	✓	✓	✓	**71.66**

HCA [31], ConfMix [37], and TSRP [35]. For a fair comparison, the methods involved in the comparison (DSGCN-ASR, DANN, ProDA, HCA, ConfMix, and TSRP) all use GCN as the backbone.

In this article, we assume that all class labels in SD participate in network training, while any class label of TD does not participate in training. The classification performance of the above methods in HT is shown in Table IV. The UH is used as the training data for the model. The PS-UDA achieves optimality in most classes but underperforms in cars and powerlines. We believe that PS-UDA does not get high-quality pseudo-labels in these classes, leading to poor classification results. Further, to evaluate the performance at a macroscopic, Fig. 11 (the 2-D space feature distribution maps) is obtained by the t-distributed stochastic neighbor embedding (t-SNE) [48] method. The proposed PS-UDA significantly outperforms other methods in classifying barren, buildings, grass, roads, and trees, which coincides with the data in Table IV. To illustrate the effectiveness of the proposed PS-UDA more straightforwardly, Fig. 12 shows the semantic segmentation map. Meanwhile, we select two representative regions to further analyze the experimental results. In Fig. 13, it is evident that the car is projected as barren. This is because superpoints confuse cars with parking lot surface points, which makes all methods not classify cars and barren well. It makes the car pseudo-labels very inaccurate. In addition, the parking lot region, which has similar spectral features to the road, is labeled as barren in the UH dataset. This also contributes to the poor prediction results of the network in barren and road classes. Furthermore, the proposed PS-UDA achieves a better advantage in the classes of building, grass, and tree. As shown in Fig. 14, PS-UDA classifies building between the tree clusters significantly better than other methods. In addition, the proposed method has significant advantages in the road and grass classes. It is worth mentioning that SqueezeSegV2 is not effective in building classification. It may be due to the fact that the interdomain differences that the method targets are different from those targeted in this article. The SqueezeSegV2 targets the dropout noise differences between synthetic and real LiDAR point clouds. The PS-UDA targets the spectral feature distribution differences between different real-scene multispectral LiDAR point clouds. Also, the method makes it difficult to consider land cover distribution, resulting in buildings being classified as cars.

On the other hand, we selected HT to train the network for UH classification. The classification performance of the above methods in UH are shown in Table V. The proposed PS-UDA achieves advantages in most classes and the overall performances. Similar to the previous description, Fig. 15 shows the advantages of the proposed method on the macroscopic level. The proposed PS-UDA outperforms other methods in the classes of barren, car, powerline, and road, which also coincides with the data in Table V. As above, the visualization of the classification in the whole scene is shown in Fig. 16, where two representative ground regions are selected for further analysis. In Fig. 17, the classification of all methods is good in barren and grass. The proposed PS-UDA is superior to the other methods in distinguishing between buildings and trees. Further, we analyze the methods through

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TABLE IV
IoU (%) OF VARIOUS FEATURES ON THE HT DATASET

Methods	Barren	Building	Car	Grass	Powerline	Road	Tree	MIoU (%)	OA (%)	F1_score (%)
K-means	0.11	1.80	2.88	0.00	0.00	0.11	1.83	0.96	2.18	1.64
SVM [45]	24.06	0.00	0.00	35.22	0.00	25.88	39.17	17.76	47.09	26.90
GCN [46]	51.13	1.01	2.33	40.00	10.45	31.66	20.97	22.51	34.45	33.29
DSGCN-ASR [5]	52.24	1.48	1.94	39.05	1.15	32.25	30.90	22.72	40.49	33.58
DANN [47]	52.31	1.26	0.49	41.33	1.65	37.31	31.18	23.65	41.23	33.68
SqueezeSeg V2 [6]	53.13	0.73	4.26	36.08	10.47	33.23	35.47	24.62	43.45	34.03
ProDA [33]	76.26	28.49	0.84	65.48	16.29	42.39	75.41	43.59	76.92	57.01
HCA [31]	78.93	55.83	12.57	66.48	18.30	70.08	35.08	48.18	78.67	63.14
ConfMix [37]	79.72	73.04	20.54	58.66	**41.32**	60.07	36.54	52.84	77.61	68.60
TSRP [35]	82.72	51.86	**22.30**	73.54	34.06	55.57	88.48	58.36	87.32	70.89
PS-UDA (ours)	**83.49**	**78.61**	16.90	**76.55**	35.25	**70.23**	**90.70**	**64.52**	**89.18**	**71.66**

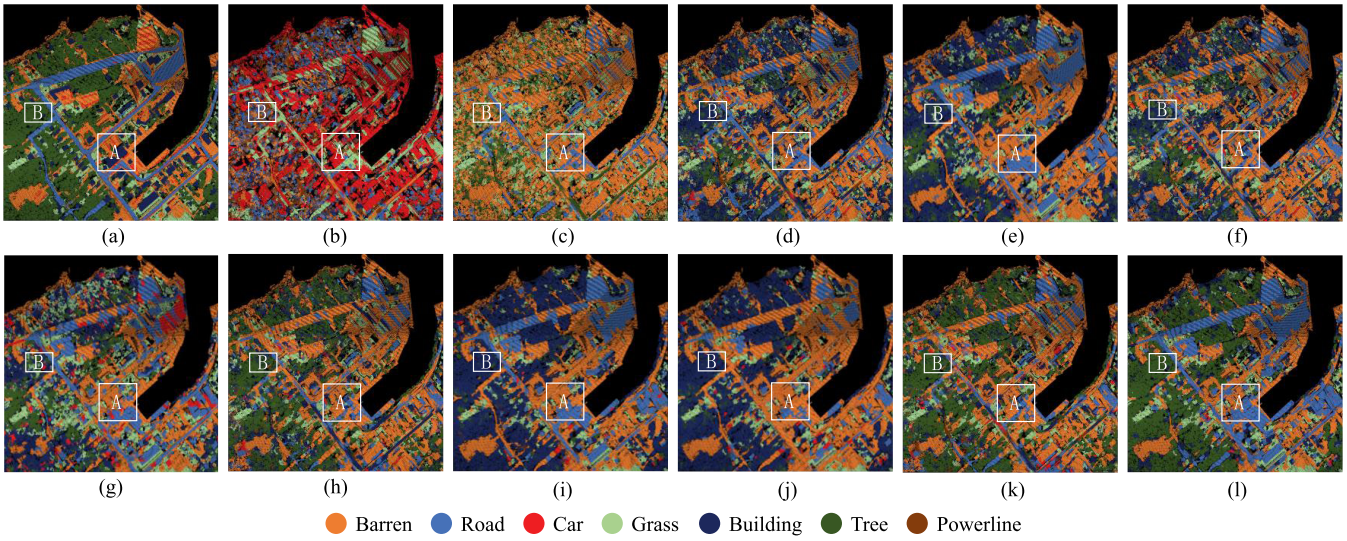


Fig. 12. Qualitative comparison of semantic segmentation maps for HT dataset. (a) Ground truth. (b) K-means. (c) SVM. (d) GCN. (e) DSGCN-ASR. (f) DANN. (g) SqueezeSeg V2. (h) ProDA. (i) HCA. (j) ConfMix. (k) TSRP. (l) PS-UDA (ours).

a 3-D perspective. As shown in Fig. 18, the proposed PS-UDA performs well in the classes of building, tree, and barren, and preserves the boundary between barren and buildings well. It is worth pointing out that only the proposed PS-UDA classifies buildings correctly in this region.

In summary, as shown in Table VI, the proposed PS-UDA can achieve the best results in most classes with a small number of parameters and low computational complexity. When the target scene is HT, the classification accuracy is the highest among the five classes of land cover. When the target scene is UH, the classification accuracy is the highest among the four classes of land cover. The proposed PS-UDA achieves the best results in all overall evaluation indicators (MIoU, OA, $F1$_score). This is due to the fact that the proposed method pre-aligns the features of the two scenes and uses Shannon entropy to constrain the pseudo-labels. The above experimental results prove the fine cross-scene classification ability of the proposed PS-UDA. It is worth noting that for small object classes such as car and powerline, all methods do not perform well. This is due to the limitation of computational resources, the point cloud containing millions of points is segmented into 8000 superpoints, resulting in each superpoint cluster being too large, leading to the confusion of small objects with their neighbors. The reasons for this are detailed in our previous work [43]. In addition, since parking lots are labeled as barren in the two datasets, and the spectral features of parking lots and roads are very similar, roads and barren are often indistinguishable during cross-scene classification. It is worth mentioning that the current state-of-the-art multispectral point cloud classification method (DSGCN-ASR) does not perform well on cross-domain classification tasks, and the method is even inferior to GCN in some classes. This is due to the strong source-domain bias of the current domain-specific multispectral point cloud classification method.

F. Discussion

Although the proposed PS-UDA is able to achieve better performance, it still has room for improvement. Firstly,

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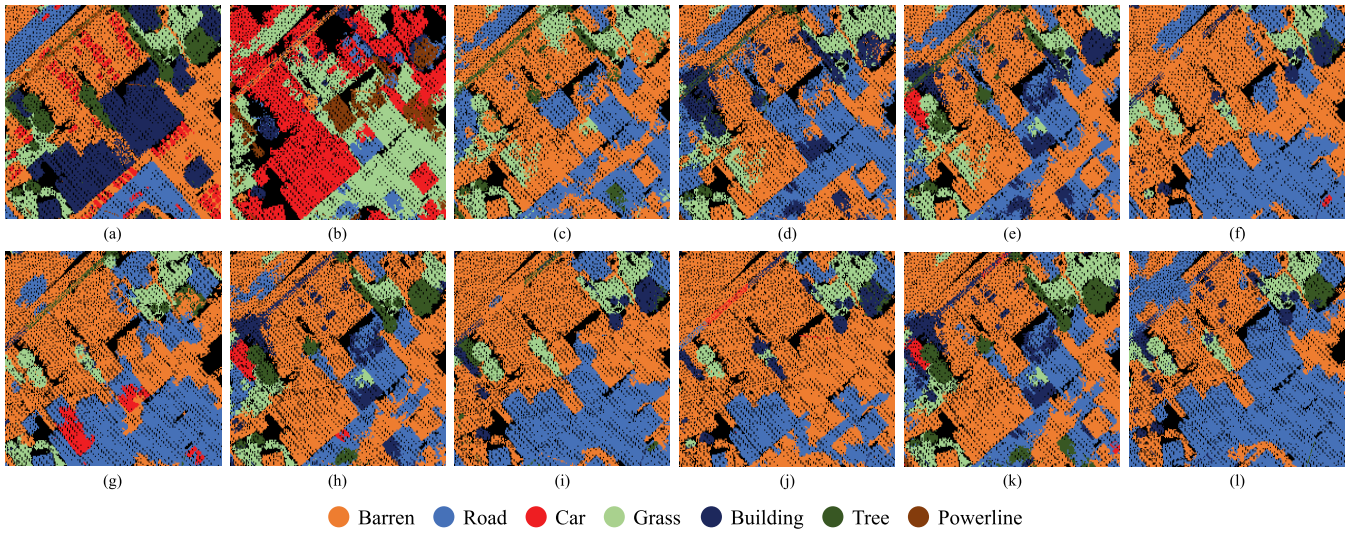


Fig. 13. Region A of the classified point cloud of the HT data set. (a) Ground truth. (b) K-means. (c) SVM. (d) GCN. (e) DSGCN-ASR. (f) DANN. (g) SqueezeSeg V2. (h) ProDA. (i) HCA. (j) ConfMix. (k) TSRP. (l) PS-UDA (ours).

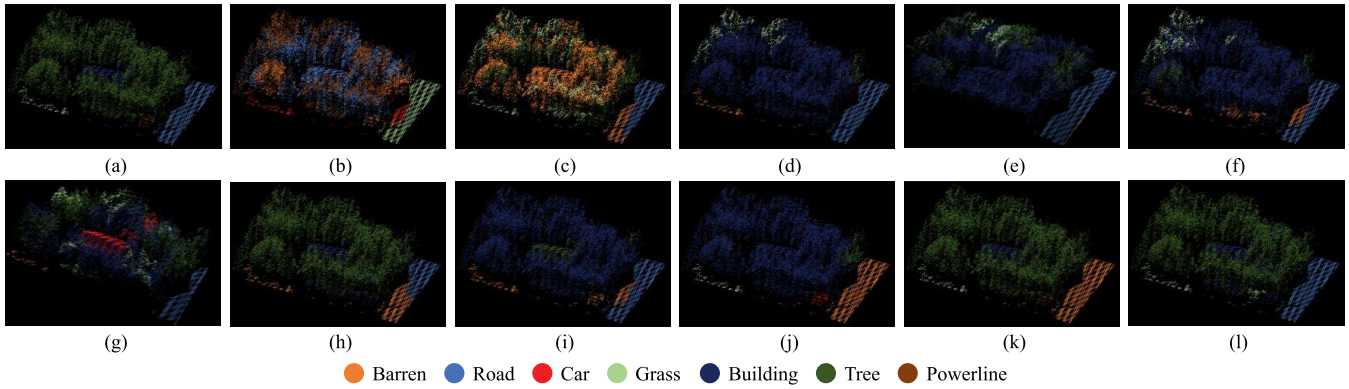


Fig. 14. Three-dimensional view of region B of the classified point cloud of the HT data set. (a) Ground truth. (b) K-means. (c) SVM. (d) GCN. (e) DSGCN-ASR (f) DANN. (g) SqueezeSeg V2. (h) ProDA. (i) HCA. (j) ConfMix. (k) TSRP. (l) PS-UDA (ours).

TABLE V
IOU (%) OF VARIOUS FEATURES ON THE UH DATASET

Methods	Barren	Building	Car	Grass	Powerline	Road	Tree	MIoU (%)	OA (%)	F1_score (%)
K-means	11.25	17.36	0.00	0.00	0.12	38.33	0.00	9.58	17.68	13.18
SVM [45]	31.75	11.30	0.00	6.69	0.00	0.10	45.07	13.56	43.13	20.48
GCN [46]	18.44	3.60	3.85	47.92	8.08	12.76	46.50	20.16	42.86	30.19
DSGCN-ASR [5]	21.23	3.34	4.01	47.69	7.12	16.16	53.33	21.84	47.12	31.43
DANN [47]	22.75	3.97	2.85	47.45	6.18	11.28	54.76	21.32	47.87	31.16
SqueezeSeg V2 [6]	29.32	4.04	2.99	50.64	15.85	15.21	54.59	24.66	49.86	33.27
ProDA [33]	26.59	56.44	28.03	70.61	14.21	20.25	80.04	42.31	70.16	55.46
HCA [31]	35.08	**67.88**	23.14	75.86	23.71	12.03	**87.56**	46.47	75.39	58.55
ConfMix [37]	29.94	65.81	18.52	**77.58**	26.64	6.92	85.22	44.38	76.04	55.87
TSRP [35]	35.34	62.27	31.36	76.70	28.95	32.34	78.51	49.35	74.82	63.59
PS-UDA (ours)	**46.12**	66.35	**39.55**	71.31	**29.41**	**59.32**	80.57	**56.16**	**76.39**	**68.01**

SD features can be further mined, such as spatial geometric features of superpoints. In this article, we focus on constraining pseudo-labels and pre-aligning features to compensate for the shortcomings of GCN. However, theoretically, designing a feature extractor specifically can fully explore the information of SD and help extract domain invariant features. In terms of

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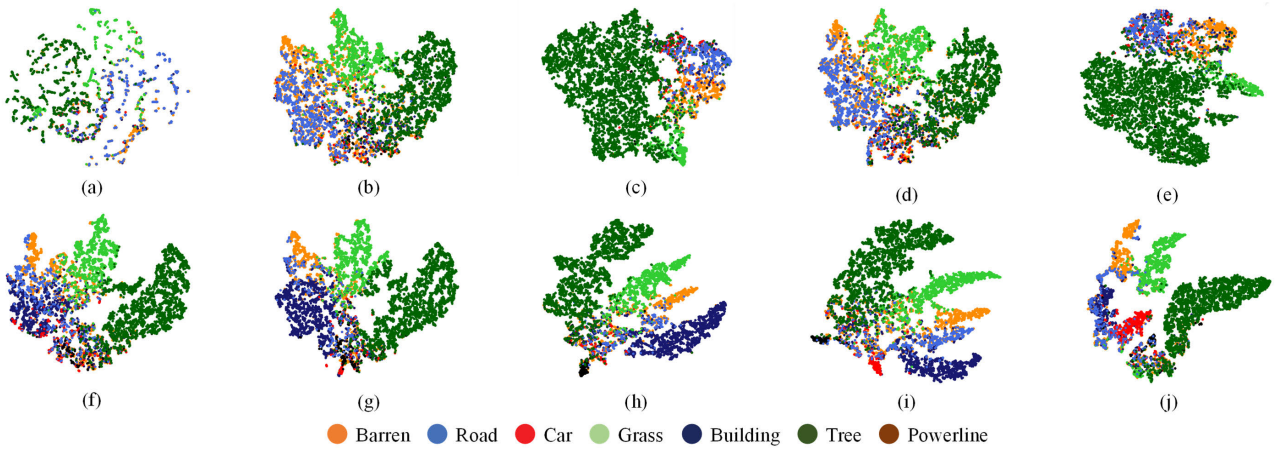


Fig. 15. Visualization of the unsupervised DA using t-SNE (The source domain is HT, and the target domain is UH). (a) SVM. (b) GCN. (c) DSGCN-ASR. (d) DANN. (e) SqueezeSeg V2. (f) ProDA. (g) HCA. (h) ConfMix. (i) TSRP. (j) PS-UDA (ours).

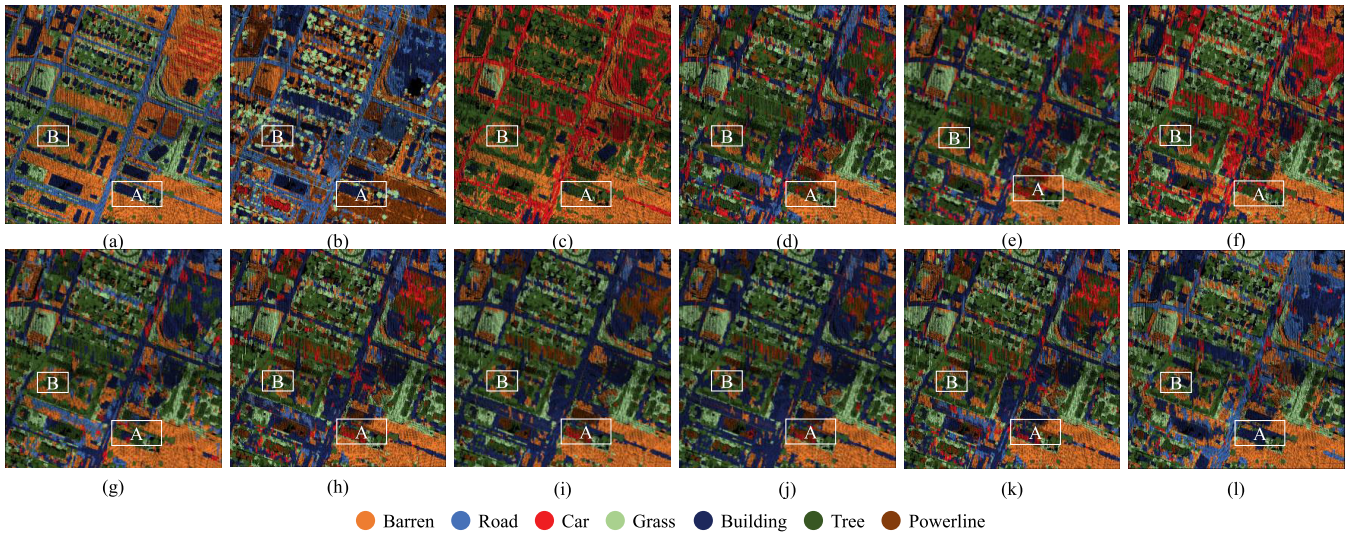


Fig. 16. Qualitative comparison of semantic segmentation maps for the UH dataset. (a) Ground truth. (b) K-means. (c) SVM. (d) GCN. (e) DSGCN-ASR. (f) DANN. (g) SqueezeSeg V2. (h) ProDA. (i) HCA. (j) ConfMix. (k) TSRP. (l) PS-UDA (ours).

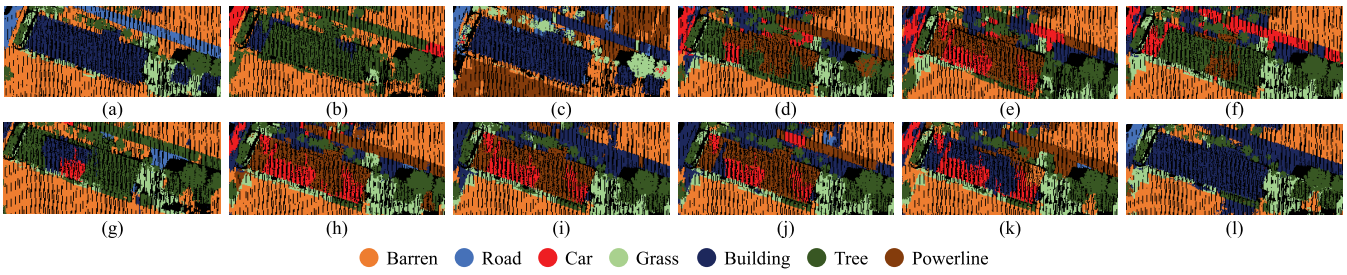


Fig. 17. Regionm A of the classified point cloud of the UH dataset. (a) Ground truth. (b) K-means. (c) SVM. (d) GCN. (e) DSGCN-ASR. (f) DANN. (g) SqueezeSeg V2. (h) ProDA. (i) HCA. (j) ConfMix. (k) TSRP. (l) PS-UDA (ours).

TABLE VI
PARAMETERS AND COMPUTATIONAL COMPLEXITY OF THE PROPOSED METHOD

Parameters	Computational complexity
250 K	1.632 GFLOPs

features, the experimental results in this article also demonstrate that intraclass feature differences in SD (IFDS) and interclass feature differences in TD (IFDT) directly affect the performance of cross-scene classification tasks. Therefore, it is necessary to design a cross-scene classification network that can focus on IFDS and IFDT to enhance the performance. Meanwhile, the limitation of computational resources forced this article to use the method of superpoint segmentation for data preprocessing. However, as mentioned earlier, superpoint segmentation degrades network performance. Due to the fact that the superpoint features used in this article are feature

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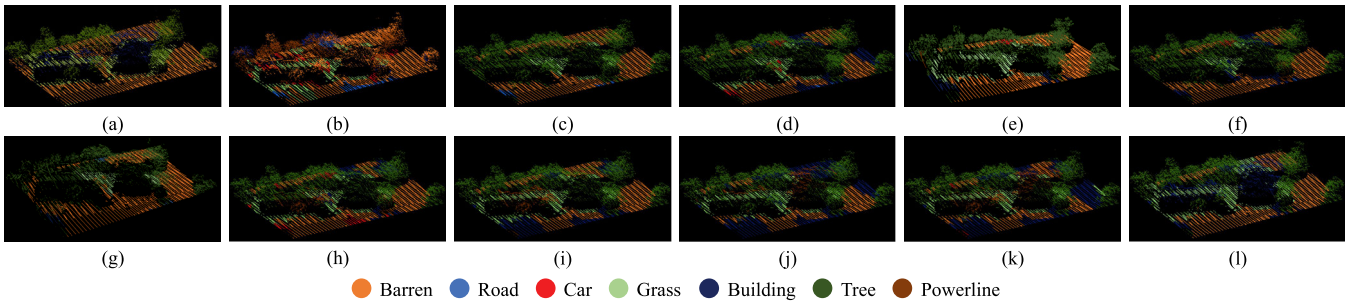


Fig. 18. Three-dimensional view of region B of the classified point cloud of the UH dataset. (a) Ground truth. (b) K-means. (c) SVM. (d) GCN. (e) DSGCN-ASR. (f) DANN. (g) SqueezeSeg V2. (h) ProDA. (i) HCA. (j) ConfMix. (k) TSRP. (l) PS-UDA (ours).

averages, there is no re-characterization of the superpoint features. The errors in the superpoint segmentation also exacerbate the confusion of the spectral features. Therefore, there is a need to develop better solutions to alleviate the computing resource requirements of point cloud data for large-scale remote sensing scenes. Moreover, for the pseudo-labels, the proposed method is selected only based on the confidence level. However, whether the confidence level is trusted is still outstanding. Therefore, there is a need to design a new pseudo-labels selection strategy to make pseudo-labels more trustworthy.

V. Conclusion

In this article, we propose a cross-scene multispectral point cloud classification method called PS-UDA based on pseudo-label, which only applies source scene labels to train the network to classify unlabeled target scene multispectral point clouds and mitigate the effects of spectral shift in the different scene MS-LiDAR point clouds. Specifically, we employ L_2 paradigm constraint and Laplace matrix to pre-align the features of the two scenes, and further employ Shannon entropy to constrain the pseudo-labels to ensure the effectiveness of subsequent training. In addition, before the second stage of training, we apply the adjacency matrix of the target scene to augment its features to fully utilize the information of the target scene. Ablation and comparison experiments show the superiority of the method for cross-scene classification of MS-LiDAR point clouds. In the future, a specialized feature extractor can be designed to fully extract the features of the source scene. It is also possible to design a network to mitigate the effects of IFDS and IFDT. In addition, we will also investigate how to optimize fixed pseudo-label thresholds to adaptive parameters.

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